

A 6–33-GHz Half-Nanosecond True-Time Delay Line With Gain Compensation for Wideband Large-Scale Antenna Array

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Abstract—This article presents a path-selecting true-time delay (TTD) circuit for wideband large-scale antenna arrays. An in-depth system-level analysis of wideband arrays is conducted to demonstrate the advantage of hybrid array system steered by both phase shifters and TTDs. The corresponding TTD design requirements are derived to enable proper operation of such arrays. The proposed T-coil peaking amplifiers are incorporated to compensate for the steep frequency-scaling losses introduced by the large number of delay lines, thereby ensuring a flat gain across an ultra-wide bandwidth. A detailed design methodology for the T-coil network is presented, and multi-coil coupled floorplan are utilized to facilitate compact layout implementation. In the high-delay path, field-reinforced floorplan are adopted for artificial transmission lines to enhance area efficiency, while in the low-delay path, distributed loss-shaping networks are introduced to mitigate loss variations across different delay states. The proposed TTD circuit is fabricated in 65-nm bulk CMOS, occupying a compact chip area of 1.6×1.2 mm². It achieves a measured delay range of half a nanosecond with 6-bit resolution across 6–33 GHz. The losses from delay lines are effectively compensated, yielding in an overall gain of 0 dB. The combination of wide bandwidth, large delay range, and low loss makes the proposed TTD circuit well-suited for wideband applications.

Index Terms—True-time delay (TTD), phased array, hybrid array, T-coil peaking, wideband, CMOS.

I. INTRODUCTION

BEAMFORMING in antenna arrays has been widely adopted for wireless links, such as satellite and 5G communications [1], [2], [3], [4]. However, the antenna arrays steered by phase shifters suffer from beam squinting issues, making them unsuitable for wideband applications. The increasing demands for high-data-rate communication necessitates massive antennas and wideband operation, promoting further research on true-time delay (TTD) elements [5], [6].

The architectures of TTD circuit include g_m -C [7], [8], [9], trombone-line [10], [11], [12], [13], internal switched [14], [15], [16] and path-selecting circuits [17], [18], [19], [20],

[21]. The g_m -C circuit is implemented with active all-pass networks (APNs), featuring compact size due to the absence of on-chip inductors. However, transistor parasitic capacitances limit its application at millimeter-wave frequencies. The work in [8] demonstrates a g_m -C circuit that achieves a 1.45 ns delay range within a compact 0.55 mm² area, but its operating bandwidth is confined to 0.1–2 GHz. In trombone-line architecture, the length of delay lines is switched by amplifiers with high input and output impedances. These amplifiers provide gain compensation, and their parasitic capacitances are absorbed by the transmission lines, enabling a wideband operation. The primary drawback of the trombone-line architecture is its limited number of control bits. The finite input and output impedance of the amplifiers can introduce discontinuities in the delay lines. Cascading additional delay units degrades the delay performance. Most reported trombone-line circuits [10], [11], [12], [13] implement only 3-bit control. The internal switched circuits utilize high-order passive artificial transmission lines (ATLs) or APNs with bypass switches. The embedded switches enable a compact floorplan, but delay performance is sensitive to the on-resistance of the switches. Reference [15] proposes a moderate-loss internal switched structure for wideband operation albeit with increased insertion loss.

The path-selecting architecture is a promising solution for achieving large delays at high frequencies. Single-pole double-throw (SPDT) or double-pole double-throw (DPDT) switches are utilized to select the signal paths. The path of high delay (i.e., delay path) consists of multiple cascaded delay units, such as ATLs and APNs, whereas the path of low delay (i.e., reference path) is realized with a short line. The path-selecting architecture exhibits lower insertion loss compared to the internal switched architecture. The former requires only two switches per delay state, whereas the latter places a switch in every delay unit [22]. Designing path-selecting TTD circuits involves two major challenges: loss variation across delay states and loss variation over frequency. The former arises primarily from the difference in path lengths between high- and low-delay states, leading to mismatch in insertion loss. Introducing additional loss in the low-delay path is a common approach to mitigate this variation. Reference [23] utilizes parallel resistors, reducing loss variation to ± 2 dB over 3–20 GHz. Reference [21] proposes asymmetrical switches for loss compensation, achieving a loss variation of ± 2.4 dB within 2–15 GHz. Nevertheless, introducing excessive loss

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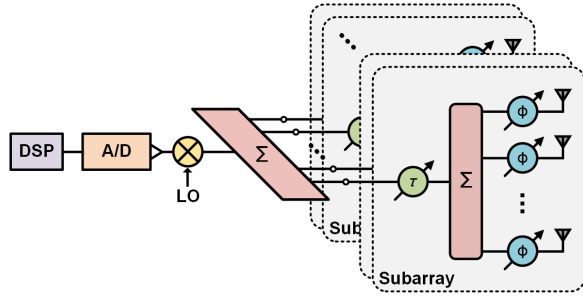


Fig. 1. Topology of analog hybrid array.

associated with these techniques can notably impair impedance matching, resulting in a restricted operational bandwidth. The second challenge, which concerns loss variation over frequency, becomes more pronounced under conditions of large delay range and wide bandwidth. The reported 315-ps delay line exhibits an insertion loss of 10 dB at 2 GHz, while reaching 36.8 dB at 15 GHz [21]. Consequently, additional amplification is required to compensate for this loss in a functional system. Off-chip amplifiers typically provide only flat in-band gain [24], [25], [26]. Although their bandwidth may be wide, the rapid decline in overall gain within the band renders them unsuitable as post-TTD gain stages for loss compensation. Therefore, on-chip amplifiers are expected to fully compensate for steep frequency-scaling loss of the delay lines, enabling a wide bandwidth.

This article proposes a 6–33-GHz half-nanosecond TTD circuit with path-selecting architecture. To enable its application in the large-scale antenna arrays and to address the above challenges, three key innovations are proposed.

- 1) An in-depth system-level analysis of hybrid arrays is conducted to highlight the necessity of TTD elements in wideband application, and the corresponding TTD design requirements are derived in Section II.
- 2) A comprehensive analysis and design methodology of the T-coil peaking amplifier are provided in Section III, ensuring a flat overall gain across an ultra-wide bandwidth. The T-coil networks are implemented using multi-coil coupled floorplan, significantly reducing the layout area.
- 3) Delay lines of low loss variation is proposed in IV, incorporating distributed loss-shaping networks to introduce frequency-dependent loss in the reference path. Additionally, the delay path of delay lines employs a field-reinforced floorplan to enhance area efficiency.

Section V depicts the measurement results and discusses the feasibility of applying the proposed chip in large-scale arrays. Section VI concludes this article.

II. SYSTEM CONSIDERATIONS AND TTD ARCHITECTURE

Implementing a timed array presents significant challenges. The required delay range of TTD elements increases with the array size, reducing their versatility within the system and complicating array integration. Furthermore, the large delay range requires substantial chip area in each channel, leading to excessive cost. To address these challenges, hybrid

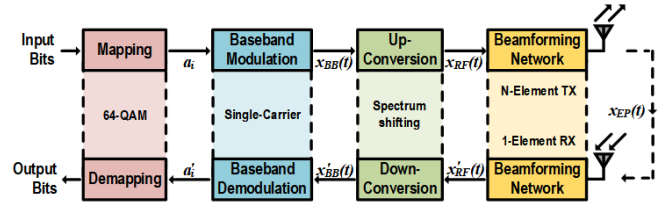


Fig. 2. Block diagram of simulation link.

array architectures are favored. Fig. 1 shows the topology of hybrid antenna array, employing a moderate number of TTD elements in wideband array systems. The array is partitioned into multiple subarrays. Phase shifters are used in each channel, while TTD elements compensate for inter-subarray time delays. Compared to the digital configuration, the analog array offers low system complexity and reduced implementation cost [27].

A. Time-Domain Model for Hybrid Array

A link is establish in Fig. 2 to investigate the impact of phasing shifting and true-time delay on the performance of a hybrid array system. Signals before beamforming is modeled as

$$x_{RF}(t) = x_{BB}(t) \cdot e^{j2\pi f_c t} = \sum_k a_k g(t - kT_s) \cdot e^{j2\pi f_c t} \quad (1)$$

where $x_{BB}(t)$ denotes the baseband signal in the time-domain, and a_k , $g(t)$, T_s denote the sequence of symbols, the time response of digital shaping filter, and symbol sampling interval, respectively. The factor $e^{j2\pi f_c t}$ accounts for the spectrum shifting to the carrier frequency, f_c . For an array consisting of N elements, partitioned into P subarrays each of size Q (i.e., $N = P \cdot Q$), the (p, q) notation is used to denote the q -th element within the p -th subarray. The required delay in each element is $(pQ + q)\tau$, where τ denotes the propagation delay between adjacent antennas. A true-time delay of $pQ\tau$ is applied in each subarray, while the residual delay is compensated by a phase shift of $2\pi f_c q\tau$ in each element. Therefore, the signal after beamforming is

$$x_{p,q}(t) = x_{RF}(t - pQ\tau) \cdot e^{-j2\pi f_c \cdot q\tau} \quad (2)$$

The radiated signal at the equiphase surface is given by

$$\begin{aligned} x_{EP}(t) &= \sum_{p=0}^{P-1} \sum_{q=0}^{Q-1} x_{p,q}(t + (pQ + q)\tau) \\ &= P \sum_{q=0}^{Q-1} \sum_k a_k g(t + q\tau - kT_s) \cdot e^{j2\pi f_c t} \end{aligned} \quad (3)$$

After demodulation and sampling at iT_s , the received symbol can be expressed as

$$\begin{aligned} a'_i &= \frac{1}{Q} \sum_{q=0}^{Q-1} \sum_k a_k g(iT_s + q\tau - kT_s) \\ &= \frac{1}{Q} \left[a_i \cdot \sum_{q=0}^{Q-1} g(q\tau) + \sum_{q=0}^{Q-1} \sum_{k \neq i} a_k g(q\tau + (i - k)T_s) \right] \end{aligned} \quad (4)$$

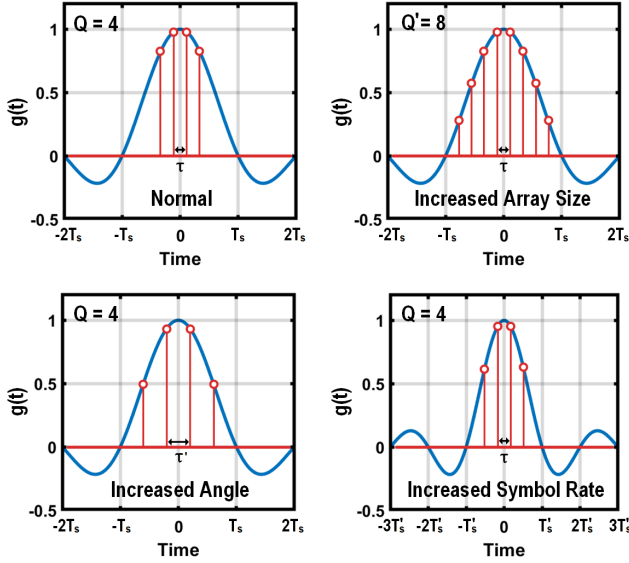


Fig. 3. Gain degradation induced by phase shifting with increased subarray size (i.e., increased Q), angle (i.e., increased τ) and symbol rate (i.e., decreased T_s). The roll-off factor of digital shaping filter is assumed to be 0.

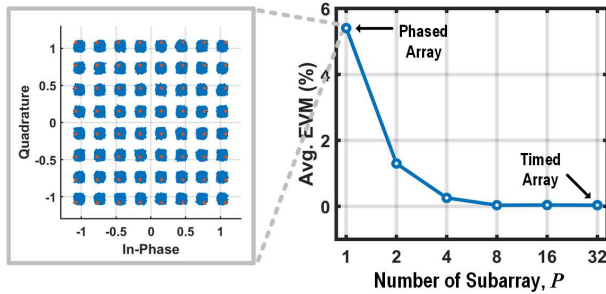


Fig. 4. Simulated average EVM of hybrid array with varying subarray number. Constellation diagram of phased array (i.e., $P = 1$) is presented.

The time response of digital shaping filter $g(t)$, where the raised cosine spectrum is mostly applied, satisfies $g(t) \leq 1$ with equality holding only at $t = 0$. For a timed array (i.e., $Q = 1$), $a'_i = a_i$ is established, indicating zero error-vector-magnitude (EVM). For a phased or hybrid array (i.e., $Q \geq 1$), a gain degradation is observed from the former item of (4). As illustrated in Fig. 3, the gain is further reduced with larger subarray size, wider scanning angle, and wider signal bandwidth. The latter term in (4) can be interpreted as additional interference noise caused by symbol overlap between antennas.

B. Determination of Subarray Number

According to the model in (4), an EVM simulation is performed to determine the required subarray number in hybrid array system. The parameters of the array are summarized in Table I. As shown in Fig. 4, the amplitude distortion and additional noise are observed in the constellation diagram, which correspond to the first and second terms in (4), respectively. For the case of a single subarray (i.e., a phased array), the average EVM is 5.5%. When the subarray number increases

TABLE I
KEY PARAMETERS OF THE ANTENNA ARRAY FOR SIMULATION

Array Size	32×32
Antenna Spacing	$0.5\lambda_c$
Scanning Range	$\pm 60^\circ$
Band Coverage	10–30 GHz
Signal Bandwidth	$0.08f_c$
Symbol Mapping	64-QAM
Modulation	Single-Carrier

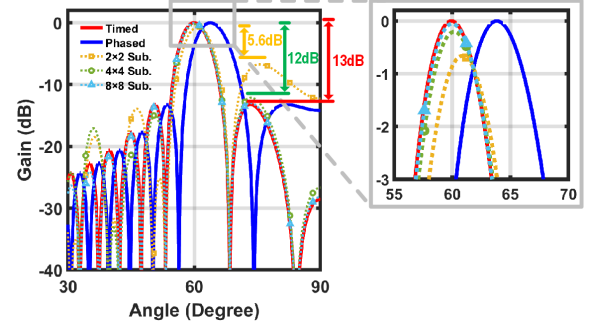


Fig. 5. Beam pattern of timed array, phased array, hybrid array with 2×2 subarrays, 4×4 subarrays, 8×8 subarrays.

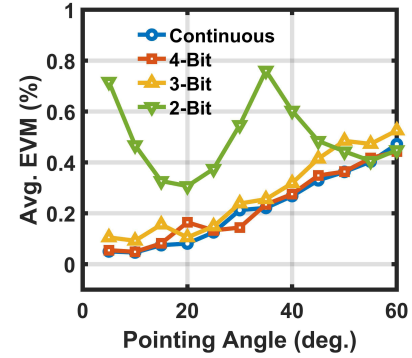


Fig. 6. Simulated average EVM with varying pointing angle and precision.

to 2×2 , the EVM drops to 1.3%, and further decreases to near zero with a 4×4 subarray configuration.

Sidelobe suppression is another critical factor in determining the subarray number. When using a 2×2 subarray, as shown in Fig. 5, the loss improves from 3 dB to 0.9 dB compared to the phased array, but sidelobe rejection decreases to only 5.6 dB. The 4×4 subarray exhibits a gain degradation of 0.2 dB and a sideband rejection of 12 dB, demonstrating performance comparable to the array steered by only TTDs. Further increasing the subarray number does not result in significant improvement. With the consideration of antenna gain, sidelobe suppression and EVM performance, a 4×4 configuration is preferred.

C. Delay Range and Precision Requirements

For an $N \times N$ uniform rectangular array (URA) with $P \times P$ subarrays, the required delay range is calculated to be

$$\tau_{req} = \frac{\sqrt{2}N(P-1)}{P} \cdot \frac{d \sin \theta_{max}}{c} \quad (5)$$

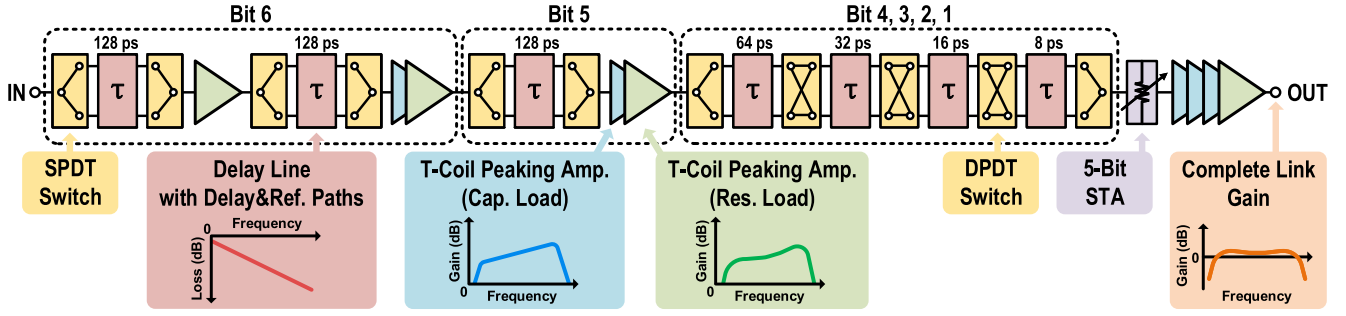


Fig. 7. Topology of proposed TTD circuit.

where d is the adjacent antenna spacing, and θ_{max} is the maximum scanning angle. The maximum delay range demanded by the array is dictated by its minimum operating frequency and is calculated to be 1.47 ns at 10 GHz. This requirement can be alleviated by cascading three TTD circuits, thereby reducing the delay range required for each TTD stage to 490 ps.

Fig. 6 reveals the impact of the delay precision on EVM. A 4-bit delay precision is sufficient, yielding an EVM comparable to that of continuously tunable delay. At higher frequencies, the requirement for delay resolution becomes more stringent. The TTD step is calculated to be 33 ps at 30 GHz.

D. TTD Architecture

Fig. 7 illustrates the topology of the proposed path-selecting TTD circuit with 6-bit precision. The delay or reference path in the delay lines is selected via SPDT or DPDT switches. To minimize loss variation across delay states, attenuators and frequency-dependent lossy networks are incorporated in the reference paths. Additionally, a 5-bit switched-type attenuator (STA) with 0.5 dB step size is employed for gain calibration.

When the delay line becomes excessively long, two major issues arise. First, the superposition of reflected signals causes significant fluctuations in gain and delay across frequencies. Second, considerable insertion loss is introduced, and its frequency-dependent nature severely limits the operational bandwidth. Both challenges can be effectively addressed by inserting amplifiers between delay lines. Serving as terminations, these amplifiers absorb reflected signals, preventing interference with other delay segments, and simultaneously compensate for the loss introduced by the delay lines. In this work, T-coil peaking networks are proposed for the matching networks of amplifiers to achieve an increased gain with frequency. T-coil networks are implemented with a capacitive load for inter-stage matching and a resistive load for output matching.

III. DESIGN OF T-COIL PEAKING AMPLIFIERS

The schematic of the proposed T-coil peaking amplifier is shown in Fig. 8. A resistive-load stage is included for output matching, while cascaded capacitive-load stages are employed

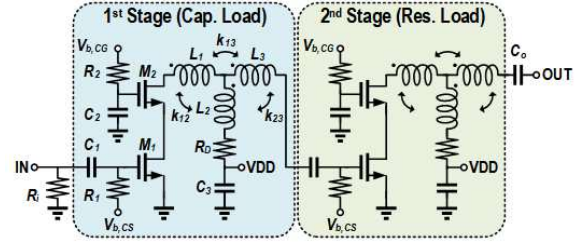


Fig. 8. Schematic of proposed T-coil peaking amplifier. The number of capacitive-load stages is determined by the required link gain.

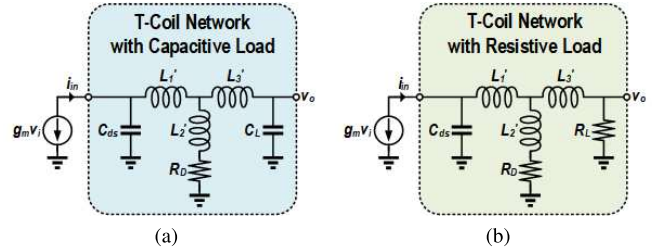


Fig. 9. Equivalent schematic of T-coil network with (a) capacitive and (b) resistive load.

to provide the required gain. Input matching of the amplifier is achieved using resistor R_i . The T-coil network, comprising three mutually coupled inductors (L_1 , L_2 , and L_3) and a resistor (R_D), is employed to realize a boosted gain response with frequency. Fig. 9 shows the equivalent schematic of the T-coil network, where the three mutually coupled inductors are modeled as three independent inductors:

$$\begin{aligned} L'_1 &= L_1 + M_{12} + M_{13} + M_{23} \\ L'_2 &= L_2 + M_{12} - M_{13} - M_{23} \\ L'_3 &= L_3 - M_{12} + M_{13} - M_{23} \end{aligned} \quad (6)$$

The output capacitance of the transistors (C_{ds}) is included in the input terminal of the T-coil network. Under the capacitive load condition, the network is loaded by the input capacitance of the subsequent stage, which can be approximated to the gate-to-source capacitance (C_{gs}) under the unidirectional assumption. For the resistive load case, the load corresponds to the characteristic impedance of the delay lines.

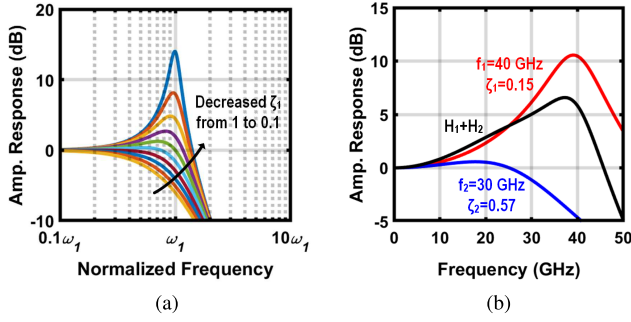


Fig. 10. Amplitude response for (a) one and (b) two pair of conjugated poles.

A. Capacitive Load for Inter-Stage Matching

The transimpedance of the capacitive loaded T-coil network is derived as follows:

$$Z_{21c} = R_D \cdot \frac{a_{1c}s + 1}{b_{4c}s^4 + b_{3c}s^3 + b_{2c}s^2 + b_{1c}s + 1} \quad (7)$$

where

$$a_{1c} = L'_2/R_D \quad (8)$$

$$b_{4c} = C_{ds}C_L (L'_1L'_2 + L'_1L'_3 + L'_2L'_3) \quad (9)$$

$$b_{3c} = R_DC_{ds}C_L (L'_1 + L'_3) \quad (10)$$

$$b_{2c} = C_{ds} (L'_1 + L'_2) + C_L (L'_2 + L'_3) \quad (11)$$

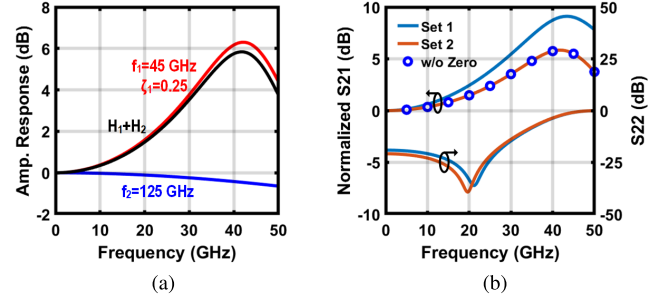
$$b_{1c} = R_D (C_{ds} + C_L) \quad (12)$$

Real poles in the transfer function result in a monotonically decreasing frequency response, which is undesirable for this design. In contrast, as shown in Fig. 10(a), a pair of complex-conjugate poles can generate a frequency response that increases with frequency. Accordingly, the transfer function is designed to include two pairs of complex-conjugate poles. The transfer function with a pair of complex-conjugate poles (p_1 and p_1^*) can be expressed as

$$\begin{aligned} H_1(s) &= \frac{1}{(s - p_1)(s - p_1^*)} \\ &= \frac{1}{\omega_1^2 \left(\frac{s^2}{\omega_1^2} + 2\zeta_1 \frac{s}{\omega_1} + 1 \right)} \end{aligned} \quad (13)$$

where $\omega_1 = |p_1|$ and $\zeta_1 = -\text{Re}(p_1)/|p_1|$. In Fig. 10(a), replacing variables p_1 and p_1^* with ω_1 and ζ_1 provides a more intuitive representation of the relationship between the variables and their corresponding amplitude responses. When ζ_1 decreases from 1 to 0.1, $|H_1(s)|$ exhibits an increasing peak, and the peak frequency gradually approaches ω_1 . Nonetheless, the peaking response may not effectively match the loss trend of the delay line. Thus another pair of poles are introduced to control the gain compensation rate. As depicted in Fig. 10(b), by carefully designing the values of ω_1 , ζ_1 , ω_2 and ζ_2 , a smoothly increasing amplitude response can be achieved to counteract passive losses, ultimately leading to a flat overall gain.

Based on the above analysis, a target transfer function is derived. The subsequent step is to synthesize the circuit

Fig. 11. (a) Amplitude response for one real pole and one pair of conjugated poles. (b) Solved S_{21} and S_{22} with solution set 1 and set 2.

corresponding to this function. The denominator polynomial of the transfer function can be represented by

$$\begin{aligned} D_c(s) &= \left(\frac{s^2}{\omega_1^2} + 2\zeta_1 \frac{s}{\omega_1} + 1 \right) \left(\frac{s^2}{\omega_2^2} + 2\zeta_2 \frac{s}{\omega_2} + 1 \right) \\ &= b_{4c}s^4 + b_{3c}s^3 + b_{2c}s^2 + b_{1c}s + 1 \end{aligned} \quad (14)$$

where

$$b_{4c} = \frac{1}{\omega_1^2 \omega_2^2} \quad (15)$$

$$b_{3c} = \frac{2}{\omega_1 \omega_2} \left(\frac{\zeta_1}{\omega_2} + \frac{\zeta_2}{\omega_1} \right) \quad (16)$$

$$b_{2c} = \frac{1}{\omega_1^2} + \frac{1}{\omega_2^2} + \frac{4\zeta_1\zeta_2}{\omega_1\omega_2} \quad (17)$$

$$b_{1c} = 2 \left(\frac{\zeta_1}{\omega_1} + \frac{\zeta_2}{\omega_2} \right) \quad (18)$$

By substituting (15)–(18) into (9)–(12), and solving the four equations for the four variables (L'_1 , L'_2 , L'_3 , and R_D), the circuit parameters can be determined. Specifically, the R_D can be solved from (12), while (10) and (11) indicate linear relations among L'_1 , L'_2 and L'_3 . Therefore, (9) is simplified into a quadratic equation with one unknown, leading to two possible solutions to equation sets. It is worth noting that the solutions for L'_1 , L'_2 and L'_3 can be negative, and through (6), they can be transformed into feasible inductance values and coupling coefficients.

B. Resistive Load for Output Matching

The transimpedance (Z_{21r}) of the resistive loaded T-coil network in Fig. 9(b) becomes a 3-pole function:

$$Z_{21r} = \frac{R_D R_L}{R_D + R_L} \cdot \frac{a_{1r}s + 1}{b_{3r}s^3 + b_{2r}s^2 + b_{1r}s + 1} \quad (19)$$

where

$$a_{1r} = L'_2/R_D \quad (20)$$

$$b_{3r} = \frac{C_{ds}}{R_D + R_L} (L'_1L'_2 + L'_1L'_3 + L'_2L'_3) \quad (21)$$

$$b_{2r} = \frac{R_L C_{ds}}{R_D + R_L} (L'_1 + L'_2) + \frac{R_D C_{ds}}{R_D + R_L} (L'_1 + L'_3) \quad (22)$$

$$b_{1r} = \frac{1}{R_D + R_L} (L'_2 + L'_3) + \frac{R_D R_L C_{ds}}{R_D + R_L} \quad (23)$$

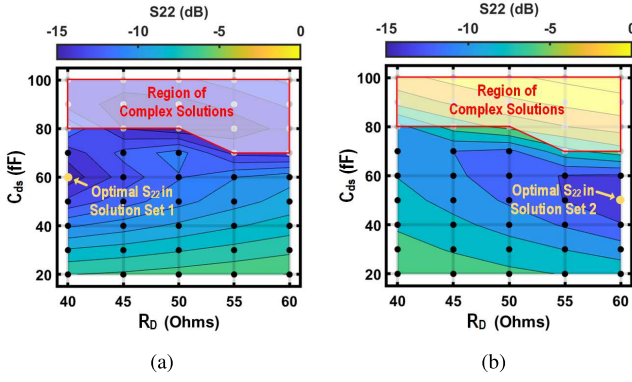


Fig. 12. Simulated S_{22} at 30 GHz for solution (a) set 1 and (b) set 2 by sweeping C_{ds} and R_D .

The function possesses one real pole and a pair of conjugated poles, with its amplitude response shown in Fig. 11(a). The real pole is designed to be much higher than the operating frequencies to achieve higher gain within the passband. Similarly, the denominator polynomial can be derived as

$$D_r(s) = \left(\frac{s^2}{\omega_1^2} + 2\zeta_1 \frac{s}{\omega_1} + 1 \right) \left(1 + \frac{s}{\omega_2} \right) = b_{3r}s^3 + b_{2r}s^2 + b_{1r}s + 1 \quad (24)$$

where

$$b_{3r} = \frac{1}{\omega_1^2 \omega_2} \quad (25)$$

$$b_{2r} = \frac{1}{\omega_1^2} + \frac{2\zeta_1}{\omega_1 \omega_2} \quad (26)$$

$$b_{1r} = \frac{2\zeta_1}{\omega_1} + \frac{1}{\omega_2} \quad (27)$$

By substituting (25)–(27) into (21)–(23), three equations are established for more than three variables, potentially leading to infinite number of solutions. In such cases, additional constraints are needed to properly refine the solutions. The output reflection coefficient is a crucial metric for evaluating the performance of the output matching network. Both the C_{ds} and R_D are considered to achieve an optimal output matching. By initially setting values for C_{ds} and R_D , a linear relationship among L'_1 , L'_2 and L'_3 can be derived from (22) and (23). Substituting this relationship into (21) results in a quadratic equation, yielding two sets of solutions.

Fig. 12 shows the calculated S_{22} at 30 GHz for the two sets of solutions by sweeping C_{ds} and R_D . Note that the R_D is restricted within the range of 40–60 Ω to ensure the output matching performance at low frequencies. The C_{ds} relates to the transistor size of the output stage that determines the achievable output power of the amplifier. The upper limit of C_{ds} is approximately 70 fF, as a larger value would lead to complex solutions of inductances. The optimal S_{22} is achieved when R_D is 40 Ω and C_{ds} is 60 fF in solution set 1, while R_D is 60 Ω and C_{ds} is 50 fF in solution set 2.

Fig. 11(b) illustrates the normalized S_{21} and S_{22} of the two solution sets. Both of the sets achieve a S_{22} below -10 dB across DC to 33 GHz. The solved L'_2 in set 2 is extremely small, implying an exceedingly large zero according to (20).

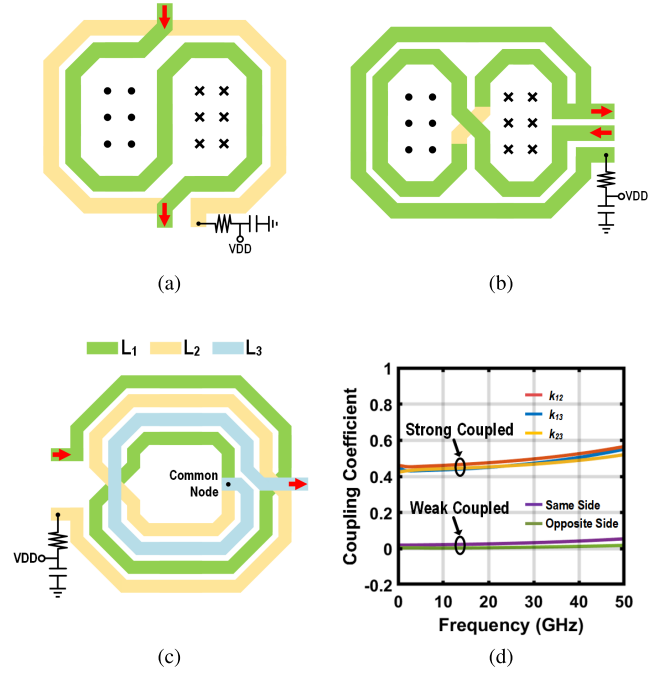


Fig. 13. Floorplan of two inductors featuring weak coupling with (a) same and (b) opposite side connection. (c) Floorplan of three inductors featuring strong coupling. (d) Simulated coupling coefficients of proposed floorplans.

TABLE II
PARAMETERS OF T-COIL NETWORK

Parameter	Cap.-Loaded		Res.-Loaded	
	Set1	Set2	Set1	Set2
M1/M2 Size (μm)	64	64	64	64
L'_1 (pH)	474	121	160	194
L'_2 (pH)	135	85	158	-4.2
L'_3 (pH)	47	400	-4.7	189
R_D (Ω)	52	52	40	60
C_{ds} (fF)	60	60	60	50
C_L (fF)	80	80	-	-
R_L (Ω)	-	-	50	50

Therefore, its corresponding S_{21} is essentially identical to that of the pure-pole system in Fig. 11(a). In contrast, the S_{21} exhibits a higher peaking gain in the solution set 1, which is preferred in this work, due to its relatively small zero.

C. Multi-Coil Coupled Floorplan

The inductors in the matching network occupy a significant portion of the layout area. In the proposed layout design, multi-coil coupled floorplan are applied to achieve a high area efficiency.

Table II summarizes the design values of T-coil peaking amplifiers. For the resistive load, the solution for L'_3 in set 1 is approximately zero. Therefore, coupling coefficient between L_1 and L_2 can be designed to be zero. Fig. 13(a) and (b) show the floorplans of two coils with low coupling for output matching. The L_1 is twisted into an S-shape for opposite

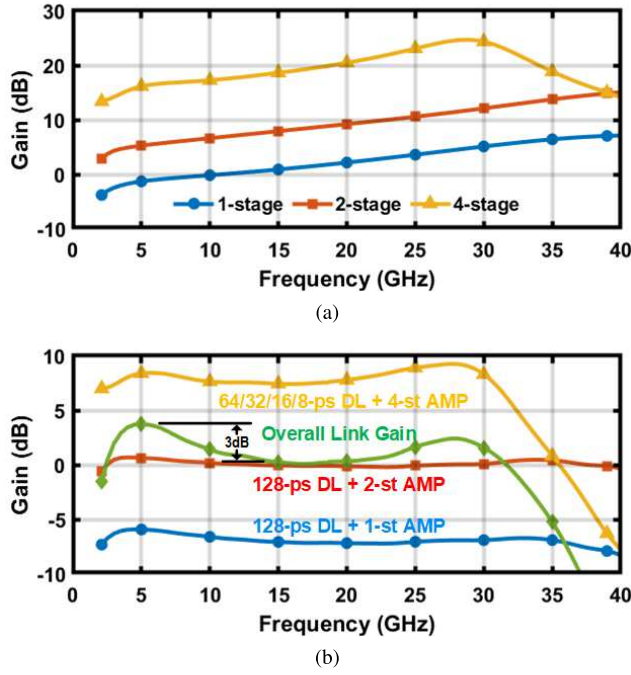


Fig. 14. (a) Simulated gain of 1-, 2- and 4-stage T-coil peaking amplifiers and (b) simulated overall gain associated with delay lines.

side connection or an 8-shape for same side connection, while L_2 is positioned on the outer loop surrounding L_1 . The magnetic fields generated by the two loops in L_1 are oppositely directed, resulting in small magnetic flux through L_2 and hence a very small coupling coefficient. The simulated coupling coefficients are shown in Fig. 13(d), which are below 0.05 for both floorplans. For the capacitive load, set 1 is selected for simple layout implementation. When all coupling coefficients are set to 0.5, L_1 , L_2 and L_3 are solved to be 198, 208 and 150 pH, respectively. Since the values of L_1 and L_2 are close, a symmetric structure can be employed. Fig. 13(c) shows the proposed floorplan of three coils with high coupling coefficients for inter-stage matching. The coupling coefficients of each pair of inductors are simulated to be around 0.45 in Fig. 13(d).

Fig. 14 illustrates the post-simulation results of 1-, 2- and 4-stage T-coil peaking amplifiers. These amplifiers achieve approximately 6-dB gain enhancement from 5 to 30 GHz, which adequately compensates for the delay line insertion loss. The complete link ultimately achieves approximately 0 dB of overall gain, with gain fluctuation maintained within 3 dB across the 2.5–31 GHz bandwidth.

D. Discussions

The T-coil peaking amplifier is characterized by f_i , ζ_i , R_D and C_{ds} , while the value of L_1 , L_2 and L_3 are obtained according to (7)–(18) and (19)–(27). The parameter f_i determines the amplifier's bandwidth, while f_i together with ζ_i governs the gain slope. R_D relates to the amplifier's DC gain, whereas C_{ds} reflects the size of the output-stage transistor and thereby dictates the deliverable output power. For a resistive load, R_D must be close to R_L to ensure proper impedance matching at low frequencies. As a result, the DC gain provided

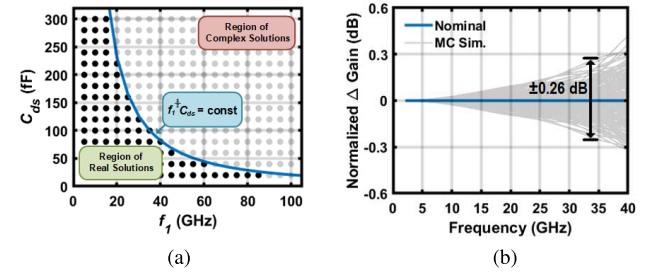


Fig. 15. (a) Trade-off between bandwidth and output power for resistive-loaded stage. (b) Monte Carlo simulations on gain variation of two-stage T-coil peaking amplifier.

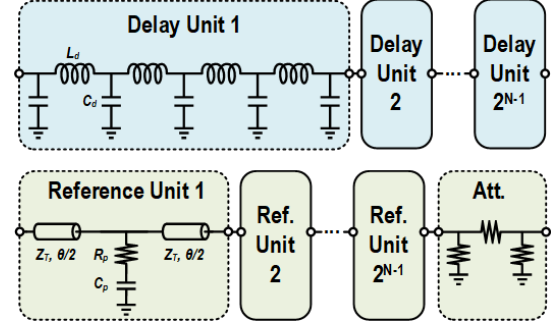


Fig. 16. Schematic of delay line composed of cascaded delay units and reference units.

by the amplifier is essentially fixed. Fig. 15(a) illustrates the feasibility of T-coil inductance solutions under various combination of f_i and C_{ds} . Note that ζ_1 is selected for a fixed gain slope. Only real solutions are physically acceptable. At the boundary between real and complex solutions, the data can be approximated by a scaling law characterized by a fixed product of $f_i^{3/2}$ and C_{ds} , revealing the fundamental trade-off between bandwidth and output power. For a capacitive load, the DC gain is derived according to the solved R_D :

$$A_{DC} = \frac{2g_m}{C_{ds} + C_L} \left(\frac{\zeta_1}{\omega_1} + \frac{\zeta_2}{\omega_2} \right) \quad (28)$$

A wider bandwidth is associated with a larger ω_1 and ω_2 , while a higher peaking gain corresponds to a smaller ζ_1 and ζ_2 , both contributing to reduced DC gain. Therefore, a trade-off is observed between the bandwidth, peaking gain and DC gain.

Monte Carlo simulations are conducted to evaluate the impact of process-induced capacitance variations on the peaking characteristic. As depicted in Fig. 15(b), the proposed two-stage T-coil peaking amplifier exhibits a gain error within ± 0.26 dB at 35 GHz, demonstrating good tolerance to process variations.

IV. DELAY LINES, SWITCHES AND STA

Tunable delay is achieved by incorporating delay lines and switches. As shown in Fig. 16, the delay lines are formed by cascading different numbers of delay and reference units.

A. Field-Reinforced Floorplan in Delay Path

In each delay unit, a 4-section ATL is implemented to achieve an 8-ps delay. The ATL is realized using multi-taper

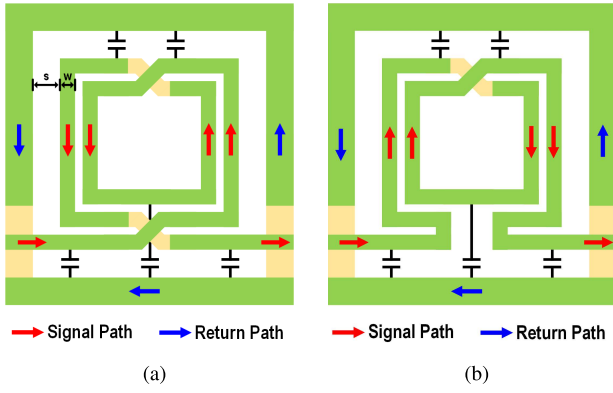


Fig. 17. (a) Proposed field-reinforced floorplan and (b) field-cancelled floorplan for delay units.

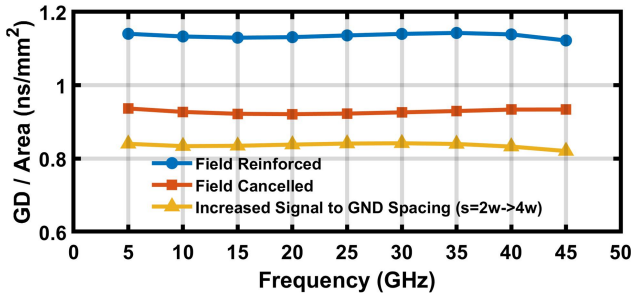


Fig. 18. Simulated area efficiency of different floorplans.

inductors. In conventional inductor layout implementations, the ground return path is typically placed at a distance from the signal path to avoid capacitive and inductive coupling that would otherwise degrade inductor performance. However, this separation results in low layout area efficiency. In ATL design, signal-ground coupling can be exploited for improved area utilization. Specifically, the capacitive coupling can be leveraged for characteristic impedance engineering, while the inductive coupling is utilized to enhance magnetic field. The proposed field-reinforced layout floorplan is depicted in Fig. 17(a). Compared to the field-cancelled floorplan in Fig. 17(b), the proposed floorplan alters the direction of current flow in the signal path to align with that in the ground loop. Consequently, the mutual coupling between the signal path and the ground loop leads to an increase in the effective inductance, allowing a larger delay. Additionally, the reduced spacing between the signal and ground paths can further enhance magnetic field, contributing to higher area efficiency. As shown in Fig. 18, where all delay lines are matched to 50Ω , the proposed floorplan achieves the maximum group delay within a given unit area.

B. Distributed Loss-Shaping Network in Reference Path

In the reference path, additional losses are intentionally introduced to match the loss in the delay path. A frequency-independent loss is introduced using a π -type attenuator at the end of the line to compensate for DC loss. A frequency-dependent loss is shaped in each reference unit by paralleling a series RC network onto the through path.

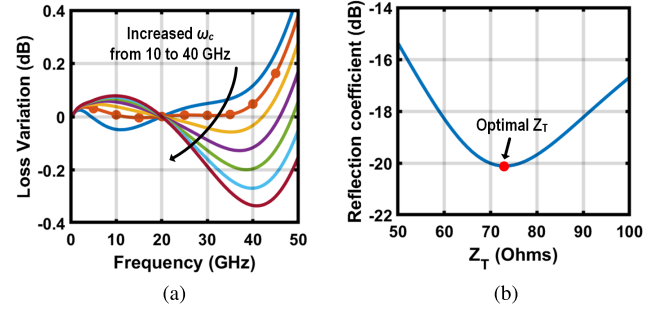


Fig. 19. (a) Loss Variation between the delay and reference units by sweeping ω_c . (b) Reflection coefficients by sweeping Z_T .

The insertion loss of the RC-based loss-shaping network can be expressed as

$$S_{21,RC} = \sqrt{\frac{1 + \left(\frac{\omega}{\omega_c}\right)^2}{1 + m \left(\frac{\omega}{\omega_c}\right)^2}} \quad (29)$$

where

$$\omega_c = \frac{1}{R_p C_p} \quad (30)$$

$$m = \left(1 + \frac{Z_0}{2R_p}\right)^2 \quad (31)$$

The gain of the network decreases from 1 to $1/m$ from DC to the frequencies where $\omega \gg \omega_c$. The frequency range where the gain decreases approximately linearly is designed to cover the operating frequency, and the rate of gain attenuation is matched to that in the delay unit. According to the gain response in the delay unit, an attenuation of 0.4 dB is expected at 20 GHz. Under this condition, the ω_c is swept in Fig. 19(a). An ω_c of 15 GHz is selected to achieve the smallest loss variation between the delay and reference units across the operating band.

In practical scenarios, the interconnections within the reference units are usually non-negligible and can be modeled as transmission lines with characteristic impedance (Z_T) and electrical length (θ). When multiple reference units are cascaded, the entire reference path exhibits distributed characteristics. The normalized transmission matrix of the T-network in the reference unit is expressed as

$$M = \begin{bmatrix} \cos \theta + jY \sin \theta & Y(\cos \theta - 1) + j \sin \theta \\ Y(\cos \theta + 1) + j \sin \theta & \cos \theta + jY \sin \theta \end{bmatrix} \quad (32)$$

where

$$Y = Z_T \frac{j\omega C_p}{2(1 + j\omega R_p C_p)} \quad (33)$$

The characteristic impedance of the proposed network (Z_c) can be calculated as [28]

$$Z_c = \frac{M_{12}Z_T}{\sqrt{M_{11}^2 - 1}} = Z_T \frac{Y(\cos \theta - 1) + j \sin \theta}{\sqrt{(\cos \theta + jY \sin \theta)^2 - 1}} \quad (34)$$

The characteristic impedance of this transmission line (Z_T) is optimized for minimum overall reflection coefficient.

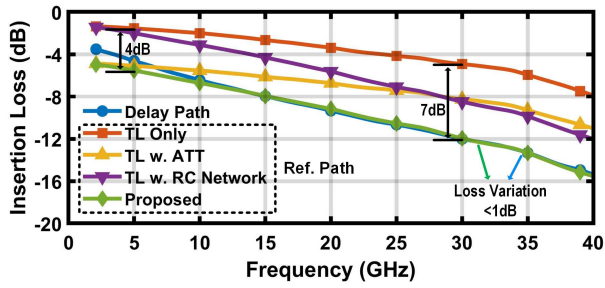


Fig. 20. Simulation result of proposed 128-ps delay line.

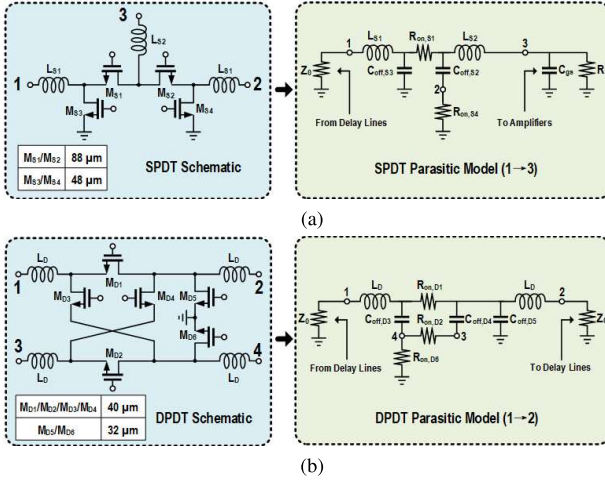


Fig. 21. Schematic of (a) SPDT and (b) DPDT, and their equivalent models.

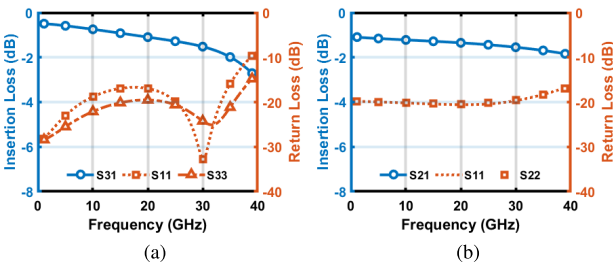


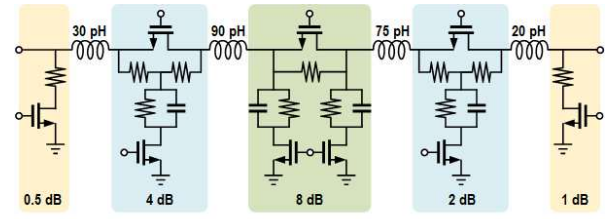
Fig. 22. Simulation results of (a) SPDT and (b) DPDT switches.

Fig. 19(b) shows the calculated reflection coefficients obtained by sweeping Z_T . Z_T is optimized to be 73Ω for the minimum reflection at the load.

The simulation results of the 128-ps delay line are shown in Fig. 20. A loss variation of 4 dB is observed at 5 GHz, while increasing to 7 dB at 30 GHz with only transmission line in the reference path. By applying both RC network and π -type attenuator, the loss variation is reduced to below 1 dB across 5–30 GHz.

C. SPDT and DPDT Switches

The schematic of the proposed SPDT and DPDT switches, along with their equivalent parasitic models, is shown in Fig. 21. In the SPDTs, the source impedance corresponds to the characteristic impedance of the transmission line, while the load impedance represents the input impedance of the amplifiers. To achieve optimal impedance matching, a resistor with a value of Z_0 is placed in parallel at the amplifier input. The input capacitances of the amplifiers, along with the series



Att.	Type	M_s (μm)	M_p (μm)	R_s (Ω)	R_p (Ω)	C_{comp} (fF)
0.5 dB	Simplified T	-	0.9	-	32	-
1 dB	Simplified T	-	2	-	39	-
2 dB	T-Type	50	4	8	125	17
4 dB	T-Type	50	8	14	58	41
8 dB	π -Type	32	8	61	70	30

Fig. 23. Schematic of proposed 5-bit STA.

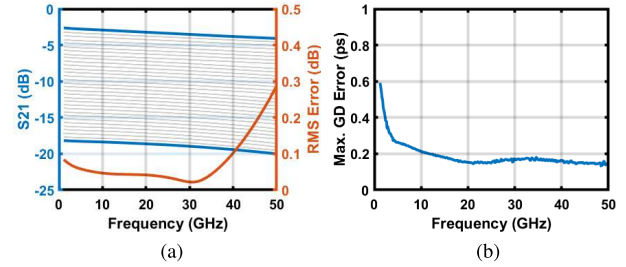
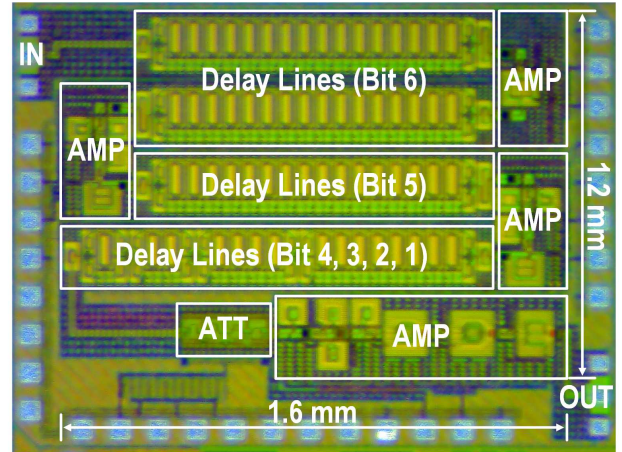
Fig. 24. Simulated (a) S_{21} and (b) group delay error of STA.

Fig. 25. Die micrograph.

inductances and off-capacitances of the transistors, form a second-order LC ladder network, achieving wideband matching. The DPDTs are used for fine-tuning adjustments. In the conducting path, a T-type matching network is implemented between the two delay lines. Due to the unidirectional nature of the signal path, the parallel transistors at the input end (nodes 1 and 3) are removed to mitigate parasitic effects in the layout. The simulated insertion losses of both SPDT and DPDT are below 2 dB at 35 GHz with return losses below -15 dB.

D. STA for Gain Calibration

The schematic of proposed STA for gain calibration is shown in Fig. 23 [29]. Simplified T-type, T-type and π -type topologies are applied to achieve 0.5-, 1-, 2-, 4- and 8- dB

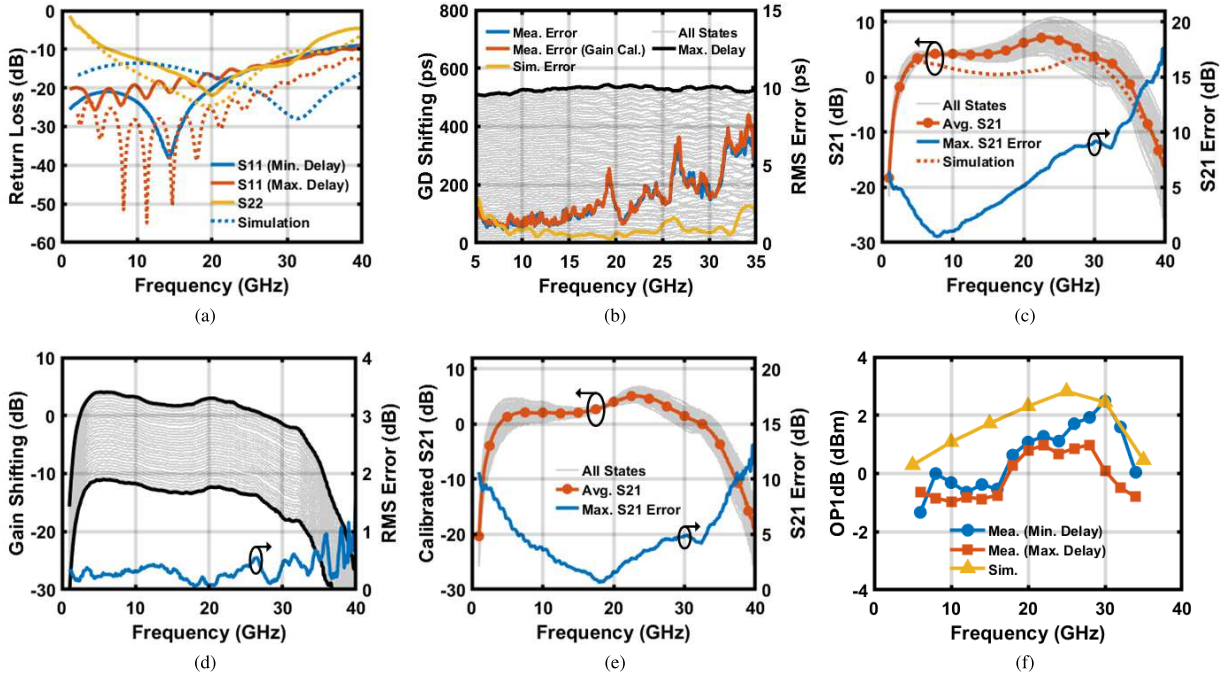


Fig. 26. Measured (a) S_{11} and S_{22} , (b) group delay shifting], (c) S_{21} , (d) gain shifting, (e) S_{21} calibrated at 18 GHz, and (f) output P_{1dB} .

attenuation, covering a total gain tuning range of 15.5 dB. Capacitive compensation is utilized to extend operating bandwidth. Fig. 24 shows the simulation result of the STA, with an RMS below 0.3 dB across DC to 50 GHz. The simulated maximum group delay error is below 0.6 ps within the band, indicating that the gain calibration introduces negligible effect on delay performance.

V. MEASUREMENT RESULTS AND DISCUSSIONS

Fabricated in 65-nm bulk CMOS, the TTD circuit occupies a total area of $1.6 \times 1.2 \text{ mm}^2$ (excluding the pads). Fig. 25 depicts the die micrograph. Drawn from a 1.8-V power supply, the circuit consume a maximal power of 180 mW. Two-port S-parameters are measured on a vector network analyzer (VNA), using two ground-signal-ground (GSG) probes.

A. Measurement Results of the TTD Circuit

Fig. 26(a) shows the measured S_{11} and S_{22} . Since the input side corresponds to the delay lines, S_{11} varies in the delay and reference states, both are below -10 dB from DC to 33 GHz. The S_{22} is below -10 dB in the 6–33 GHz band.

Fig. 26(b) shows the measured group delay of 64 states. A maximum delay of 500–540 ps is achieved across the operating band with a step of approximately 8–8.5 ps. The calculated RMS error is below 7 ps across the band.

The measured S_{21} is depicted in Fig. 26(c). A gain of approximately 0 dB is achieved at the state of maximum delay. The gain variation across all delay states is $\pm 5 \text{ dB}$. When the signal passes entirely through the delay path (maximum-delay state), the chip exhibits an almost flat gain response across the bandwidth, which is consistent with the simulation. In contrast, when the signal passes entirely through the reference path (minimum-delay state), the high-frequency gain shows

a pronounced rise. This observation indicates that the actual loss introduced by the RC network in the reference path is smaller than that predicted by simulation. As depicted in Fig. 19(a), the gain variation can be further reduced by increasing ω_c to increase the loss of RC network at high frequencies. Additionally, the gain variation can be optimized by tuning the on-chip attenuator. The gain tuning performance is shown in Fig. 26(d), achieving a 15.5 dB tuning range with RMS error below 0.5 dB. Fig. 26(e) shows the calibrated S_{21} at 18 GHz, where the gain variation is reduced to $\pm 2.5 \text{ dB}$ across 6–33 GHz. When the chip operates over a specific sub-band within the 6–33 GHz range, its gain can be calibrated in-band to achieve low gain error across the targeted frequency range.

The large signal performance is shown in Fig. 26(f). The measured output P_{1dB} is around 0 dBm.

In table III, the performance of proposed TTD circuit is summarized and compared with the prior-arts. The proposed TTD circuit achieves a delay bandwidth product of 13.6, which is the highest among the TTD circuits in the table. This is primarily attributed to the introduction of T-coil peaking amplifiers that compensate for the significant losses under large-delay and high-frequency conditions. Furthermore, the proposed TTD circuit can also provide a gain of 0 dB across a wide bandwidth with the active amplification and compensation techniques, while the reported works of trombone, internal switched and path-selecting structures exhibit considerable insertion loss, requiring off-chip amplification. This work also achieves an area efficiency (i.e., delay-to-area ratio) of 225 ps/mm² for the whole chip, and 467 ps/mm² for the delay lines only. Benefiting from the proposed compact floorplan, the highest area efficiency of delay lines is achieved among the structures employing passive delay lines.

TABLE III
PERFORMANCE COMPARISON WITH THE STATE OF THE ART

Reference	This work	[8]	[11]	[15]	[16]	[17]	[20]	[21]
Technology	65 nm CMOS	0.13 μm CMOS	0.18 μm CMOS	0.18 μm CMOS	65 nm CMOS	28 nm CMOS	65 nm CMOS	65 nm CMOS
Topology	Path-Selecting	g_m -C	Trombone	Internal Switched	Internal Switched	Path-Selecting	Path-Selecting	Path-Selecting
Delay Structure	ATL	APN	APN	APN	ATL	APN	ATL	APN
Frequency (GHz)	6–33	0.1–2	8–18	8–18	1–11	3–30	5–11	2–15
Delay Range (ps)	504	1450	109.3	125	106.7	68.5	400	315
Delay Step (ps)	8	10	15.6	3.9	4.85	4.6	1.54	5
Range $\times$ BW	13.6	2.755	1.093	1.25	1.067	1.85	0.352	4.095
Gain (dB)	0	0.6	-18.2 to -22.5	-15.2 to -23.3	-2.8 to -15.5	-7.1 to -16.7	-13 to -41*	-11.9 to -36.8
Gain Variation (dB)	± 5 (Uncal.) ± 0.25 (Cal.)	± 0.7	± 1	± 2	± 0.8	± 5	$\pm 5^*$	± 2.4
Supply Voltage (V)	1.8	1.4	3.3	Passive	Passive	Passive	Passive	Passive
DC Power (mW)	180	364	44.2	Passive	Passive	Passive	Passive	Passive
Area (mm ²)	2.24	0.55	1.89	2.0	0.35	0.34	2.46	0.72
Delay/Area (ps/mm ²)	467 ^a 225 ^b	2636 ^a	57.8 ^a	62.5 ^a	304.9 ^a	201.5 ^a	163 ^a	437.5 ^a

^a Delay Lines Only ^b Delay Lines with Active Amplification * Graphically Estimated ** Calibrated at 18 GHz

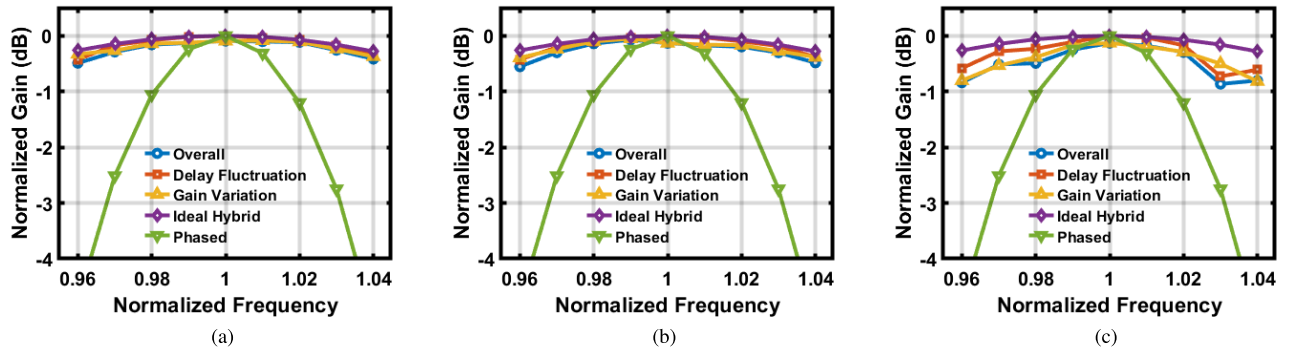


Fig. 27. Simulated signal bandwidth at (a) 10 GHz, (b) 20 GHz and (c) 30 GHz, with the consideration of delay fluctuations and gain variations.

B. Discussions

Due to the mismatching of the delay lines, the measured delay fluctuates at high frequencies, resulting in the uneven delay steps. Additionally, although the gain is calibrated at the carrier frequency, the gain still varies across the wideband range. In practical scenarios, the delay fluctuations and gain variations can lead to the deterioration of the steering performance of the array.

Simulations are performed based on the measurement results of the proposed TTD circuit. A 32×32 array with 4×4 subarrays is considered. Fig. 27 shows the simulated signal bandwidth of the array pointing to 60° . Note that three proposed TTD circuits are cascaded during simulation to meet the required delay range at low frequencies. The simulated array exhibits performance consistent with that of an ideal hybrid array at lower frequencies, owing to the relatively small delay variation. Within a signal bandwidth of $0.08f_c$, the maximum gain degradation is approximately 0.5 dB at 10 and 20 GHz. Performance degrades at 30 GHz, but the

maximum gain reduction remains below 0.9 dB, representing a significant improvement over conventional phased arrays.

VI. CONCLUSION

This article presents a path-selecting TTD circuit for wide-band large-scale antenna arrays. System considerations for large-scale arrays are conducted to determine the TTD requirements. T-coil peaking amplifiers are introduced to compensate for the losses from delay lines, contributing to a flat overall gain across ultra-wide bandwidth. In the reference path, distributed loss-shaping networks are introduced to reduce loss variations across delay states and optimize impedance matching. Both the multi-coil coupled floorplan for T-coil network and field-reinforced floorplan for delay path contribute to a high area efficiency. The proposed TTD circuit is implemented in 65-nm bulk CMOS. It achieves a measured delay range of half a nanosecond with 6-bit precision across 6–33 GHz. The losses from delay lines are compensated, providing an overall gain of 0 dB. The chip occupies a total area of $1.6 \times 1.2 \text{ mm}^2$.

The wide bandwidth, large delay range and low loss make it suitable for the application of wideband hybrid antenna arrays.

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